One gigabyte, many skills: task adapters on a 1-bit 8B model, scored on Prime Intellect environments

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Code and per-sample results: https://github.com/patrickdugan/q1-adapters · Adapters: https://huggingface.co/AlephFunk/bonsai-8b-prime-env-adapters

Draft. Every number is computed from the per-sample files in `results/` by `harness/make_tables.py` and `harness/routers.py`.

Revision note (draft 2, 2026-10-19). Draft 1 (2026-10-18) covered four environments. This draft adds five more Prime Intellect Hub environments (PubMedQA, ascii-tree, MATH-500, AIME 2024+2025, reasoning-core) on all three base models, and reruns GSM8K on its full test set (1,319) and MedMCQA on 1,000 (new §3.8). It also adds three adapters (ascii-tree, reasoning-core, PubMedQA). The abstract, §4 and §5 are revised to match. Two draft-1 statements changed with the larger samples: the GSM8K adapter's gain, "nothing measurable" at n = 200–500, is +0.020 [+0.002, +0.038] at n = 1,319; and base Q1 vs the 4-bit 1.7B on MedMCQA moved from −0.004 (n = 500) to −0.028 (n = 1,000, still not significant). §3.1–3.7 are otherwise unchanged and keep their original sample sizes.

Abstract

What is a 1-bit 8B model good for once it is given the cheapest specialization available? We score PrismML's **Bonsai 8B** (`Q1_0`, 1.16 GB) on nine **Prime Intellect Environments Hub** environments — reverse-text, GSM8K, MedMCQA, IFEval, PubMedQA, ascii-tree, MATH-500, AIME 2024+2025 and reasoning-core — using each environment's own dataset, prompt and reward code (200–1,319 items per environment). Out of the box, in non-thinking mode, it trails a 4-bit **Qwen3-8B** (`Q4_K_M`, 5.03 GB) by 4–18.5 points (within noise on reasoning-core). Against a memory-matched 4-bit Qwen3-1.7B the answer depends on reasoning load: the 1-bit 8B wins on GSM8K (+0.053, n = 1,319) and short symbolic tasks (reasoning-core +0.045) but loses on harder multi-step math (MATH-500 −0.068; AIME 0.067 vs 0.117). The reasoning-heavy environments are weak for every model at this size (AIME ≤ 0.24, reasoning-core ≤ 0.26). A rank-16 task adapter per environment (87 MB; 7–103 minutes on one laptop GPU; trained from the environment's own rewards or gold) adds **+0.70** and **+0.92** on two transformation skills neither base model has (reverse-text 0.82 and ascii-tree 0.98, vs 0.21 and 0.23 for the 4-bit 8B). It adds +0.04–0.07 on MedMCQA, PubMedQA and reasoning-core; on the last two that brings the 1-bit model level with or above the base 4-bit 8B. It adds +0.02 on GSM8K, and nothing on MATH-500. A **72k-parameter recursive router (TRM)** picks the right adapter 99.8% of the time from the prompt, and the routed four-skill library scores a macro 0.693 against 0.598 for the 4-bit 8B — in 2.4 GiB of VRAM instead of 5.2 GiB — though on the three tasks the 4-bit model can already do, it stays ahead. A lexical lattice router (LDT) used as a hard filter hurts, reproducing the TRM/LDT hybrid result. The 1-bit model decodes 2.25× faster than the 4-bit 8B on the same GPU. The public Qwen3-8B adapters we found do not help the 4-bit model under the environments' contracts. On Qwen's official `Q4_K_M` GGUF they collapse (MetaMathQA: GSM8K 0.98 → 0.37) because that file is built from an **AWQ-rescaled checkpoint** — its norm weights differ from the published model's by 20–55% — so the adapters land in the wrong weight basis; on a `Q4_K_M` built from the exact checkpoint they run as their authors report (0.86) but still lose to the base, through output-format mismatch, a missing stop behaviour, or a plain trade-off. Adapters trained on the environment's rewards against Bonsai's exact served weights avoid both problems by construction. The price of 1 bit: adapters cannot be merged into the weights, so an active adapter roughly halves served throughput.

1. Question

A 4-bit 8B model is the usual "runs on a laptop" configuration: about 5 GB of weights. PrismML's **Bonsai 8B** has end-to-end 1-bit weights — embeddings, attention, MLP and LM head — on the Qwen3-8B dense architecture: `Q1_0` stores one sign bit per weight plus one FP16 scale per group of 128 (1.125 bits/weight, 1.16 GB). PrismML's model card reports GSM8K 88 and IFEval 79.8 (EvalScope with vLLM on an H100; license Apache-2.0). A 1-bit model is weaker out of the box, so the question is not "is Q1 as good as Q4" but **what Q1 buys once you add the cheapest specialization available — a small task LoRA — and how that compares with the 4-bit model plus the adapters people already publish.**

We answer it on nine public **Prime Intellect Environments Hub** environments (four in §3.1–3.7, five more in §3.8), using each environment's own dataset, prompt and reward code, so every number below is the score the Hub environment would assign.

2. Setup

Models. - *Q1*: `prism-ml/Bonsai-8B-gguf`, `Bonsai-8B-Q1_0.gguf` (1,158,654,496 bytes), served by PrismML's llama.cpp fork (build 10743). - *Q4*: `Qwen/Qwen3-8B-GGUF`, `Qwen3-8B-Q4_K_M.gguf` (5,027,783,488 bytes), served by the same binary. - *Q4, memory-matched*: `unsloth/Qwen3-1.7B-GGUF`, `Qwen3-1.7B-Q4_K_M.gguf` (1,107,409,472 bytes — the size of the 1-bit 8B), served by upstream llama.cpp b11269 (the fork's CUDA path crashed on it). - *Q8 (diagnosis only)*: `Qwen3-8B-Q8_0.gguf` (8.71 GB), to separate adapter from quantization effects. - *Q4, correct basis*: our own `Q4_K_M` (5.03 GB, the same size as the official file), converted with upstream `convert_hf_to_gguf.py` from the published `Qwen/Qwen3-8B` safetensors and quantized with the fork's `llama-quantize` — the file the 4-bit comparisons use, because the official `Q4_K_M` turned out to be a different checkpoint (§3.6). `Q4_0`, `Q5_K_M`, `Q6_K` and an f16-embedding `Q4_K_M` were built the same way for diagnosis.

Servers ran 4–6 concurrent slots with 8k context each; from the 4-bit runs on, the host prompt cache was capped at 1 GiB, and for the runs after the GPU resets (§5) the micro-batch at 128 tokens. Neither changes the model; both change only how work is batched, which can shift floating-point rounding.

Both use the same Qwen3 chat template (taken from the Bonsai checkpoint), which always closes an empty `<think>` block — i.e. non-thinking mode for every configuration — so prompts are byte-identical across models.

Environments (installed from the Hub index; scored by their own reward functions):

Hub env	eval set	score
`primeintellect/reverse-text`	last 200 of the env's 1,000 rows (first 800 = train)	LCS ratio of parsed `<reversed_text>`
`primeintellect/gsm8k`	first 200 of GSM8K test	math-verify on `#### N`
`maziyar/OpenMed_MedMCQA`	first 200 of the env's validation split	boxed-letter accuracy
`arcee-ai/ifeval`	the env's own seeded 200-row holdout	RLVR-IFEval constraint pass rate
added in §3.8:		
`primeintellect/math500`	all 500 problems of MATH-500	math-verify on the last `\boxed{}`
`primeintellect/aime2024` + `aime2025`	all 60 problems, 2 samples each	math-verify on the last `\boxed{}` (avg@2, pass@2)
`maziyar/OpenMed_PubMedQA`	first 500 of the env's shuffled PubMedQA-labeled eval split	yes / no / maybe accuracy
`primeintellect/reasoning-core`	the env's 500-row eval stream (seed 0): grammars, logic, planning, arithmetic tasks	the env's per-task scorer
`primeintellect/ascii-tree`	last 200 of the env's rows (the rest = train)	the env's weighted tree similarity (0.3 / 0.7)

Implementation notes: the three classic-API envs require `verifiers<0.4` (0.4.0 removed `vf.Environment`); `primeintellect/gsm8k` is a verifiers-v1 taskset whose `verify.py` we run in-process with math-verify's signal/process timeouts disabled (on Windows they spawn a child that cannot re-run the script and every answer scores 0) — the scoring logic is otherwise line-for-line. `math500` and the two AIME sets are handled the same way (`math500` is a v1 taskset whose package imports POSIX-only `fcntl` through `verifiers.v1`, so its prompt, dataset revision and `verify_boxed_math_answer` are mirrored in `harness/envs.py`; every env's gold answers score 1.0 through our wrappers).

Sampling. Qwen3's recommended non-thinking settings (T = 0.7, top-p 0.8, top-k 20), one sample per eval example with a per-example seed shared by all configurations, so comparisons are paired. Max new tokens: 384 (reverse-text), 768 (gsm8k, ascii-tree), 1024 (medmcqa, ifeval, pubmedqa, reasoning-core), 2048 (math500), 6144 (AIME — non-thinking solutions to AIME problems run long; at 3072 three of four smoke-test answers were cut off mid-derivation, not looping).

Adapters. - *Ours (Bonsai)*: rank-16 LoRA on all seven projections (43.6 M parameters; 87 MB as an f16 GGUF), trained with QLoRA on `prism-ml/Bonsai-8B-unpacked` in 4-bit NF4 — whose weights are an exact expansion of the 1-bit release, so the adapter applies unchanged to `Q1_0` (we verified byte-identical generations between the torch 4-bit path and the 1-bit server in prior work). Training signal comes from the environment itself: for reverse-text the env's gold reversal (Prime Intellect's own SFT warm-up shape); for gsm8k, medmcqa and ifeval, **rejection-sampling fine-tuning** — sample 4 completions per prompt for 400 train prompts from the base 1-bit model, keep up to two distinct completions the env scores as fully correct, fine-tune on them. - *Online (Qwen3-8B)*: the best-matching public LoRAs on the Hub, used as published (not trained by us): `Pradheep1647/qwen3_8b_gsm8k_unloth_trl` (GRPO on GSM8K) and `tianzl66/Qwen3-8B-MetaMathQA-50K-LoRA` for gsm8k; `vireshk/qwen3-8b-medmcqa-qlora` for MedMCQA; `tianzl66/Qwen3-8B-InstructionFollowing-LoRA` (Tulu-3 IF split) for IFEval. All are rank 16 on all seven projections — the same shape as ours. No public adapter exists for reverse-text. (The most-downloaded "GSM8K LoRA", `swagger00/Qwen3-8B-GRPO-GSM8K`, ships merged 16 GB weights, not an adapter, and was excluded.)

Serving. One llama-server per base model with every adapter loaded at start (`--lora-init-without-apply`) and switched per run through `/lora-adapters` — the deployment pattern where one small base serves many skills.

Routers. A skill library needs a router. Following the TRM/LDT split of Dugan (2026, *Trade-offs between TRM/LDT hybrids*), we compare a **TRM** — a tiny recursive classifier (72k parameters: hashed word/bigram/char-trigram features, a latent *z* and answer state *y* refined by two weight-shared 2-layer MLPs, 3 inner × 3 outer steps, deep supervision) — with an **LDT** lexical lattice (each known word carries the candidate set of envs it occurs in; a prompt's state is the meet over its words; the empty set is bottom), and the paper's hybrid control policies: soft (lattice as telemetry), hard filter, confidence arbitration, and a typed membrane that **abstains to the base model** when the TRM's proposal conflicts with the lattice. Routers are trained on 800 train prompts per env and scored on the 800 eval prompts in two views: the full prompt as the env sends it, and the user's content with the env's instruction scaffolding removed. Besides routing accuracy we report **end-to-end** score: the env score of the answer the 1-bit model actually produced under the chosen adapter (all adapter × env answers are generated, so this is a lookup, not a model of the outcome).

3. Results

3.1 The whole picture

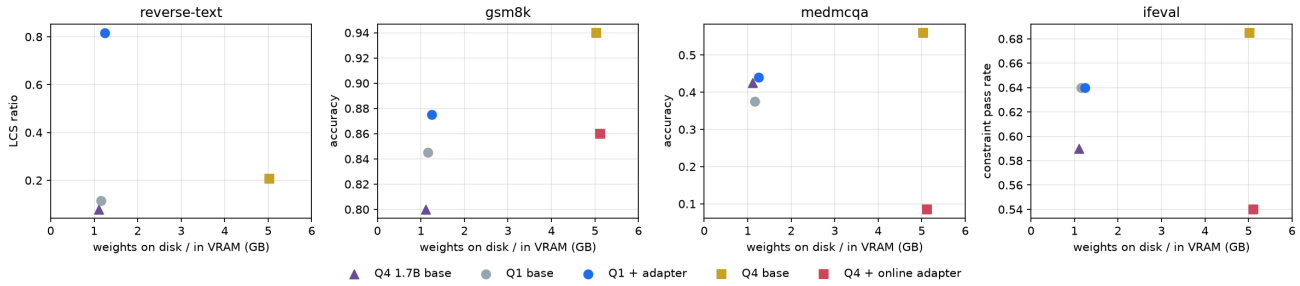
Same 200 eval items per environment for every column (paired, same seeds); weights are file sizes:

env (score)	Q4 Qwen3-1.7B (1.11 GB)	Q1 Bonsai 8B (1.16 GB)	Q1 + our adapter (+87 MB)	Q1 + online Qwen3 adapter	Q4 Qwen3-8B[3] (5.03 GB)	Q4 8B[3] + best online adapter
reverse-text (LCS)	0.079	0.115	**0.817**	—	0.208	— (none exists)
gsm8k (accuracy)	0.800	0.845	0.875	0.870	**0.940**	0.860 (MetaMathQA)
medmcqa (accuracy)	0.425	0.375	0.440	0.420	**0.560**	0.085 (MedMCQA LoRA)[4]
ifeval (pass rate)	0.590	0.640	0.640	0.630	**0.685**	0.540 (Tulu IF)
macro	0.474	0.494	**0.693**[1]	—	0.598	0.423[2]

[1] With each prompt routed to its adapter — exactly what the 72k-parameter TRM router achieves on full prompts (§3.5). [2] Using the Q4 base for reverse-text, where no public adapter exists. [3] A `Q4_K_M` we converted from the exact `Qwen/Qwen3-8B` weights the community adapters were trained on. Qwen's official `Qwen3-8B-Q4_K_M.gguf` (5.03 GB) is built from an AWQ-rescaled checkpoint and misapplies those adapters (§3.6); its base scores are 0.178 / 0.945 / 0.515 / 0.680 on these items. [4] Right letter 58.5% of the time, but in the adapter's own bare-letter format, which the env's `\boxed{}` parser rejects (§3.6).

GSM8K and MedMCQA, extended to 500 items (paired bootstrap 95% CIs):

env	Q4 1.7B	Q1 base	Q1 + our adapter	Q4 8B (official file)	Q1 + adapter vs Q1	Q1 vs Q4 1.7B	Q1 + adapter vs Q4 1.7B	Q1 + adapter vs Q4 8B
gsm8k	0.802	0.854	0.868	0.938	+0.014 [−0.014, +0.044]	**+0.052** [+0.014, +0.088]	**+0.066** [+0.032, +0.102]	**−0.070** [−0.096, −0.046]
medmcqa	0.392	0.388	0.450	0.518	**+0.062** [+0.018, +0.108]	−0.004 [−0.048, +0.044]	**+0.058** [+0.010, +0.106]	**−0.068** [−0.122, −0.014]



Score against weight size, 200-item values. On MedMCQA the 200-item sample puts the 4-bit 1.7B above the 1-bit base (0.425 vs 0.375); on 500 items they tie (0.392 vs 0.388).

Three regimes, read off the table:

- **At equal memory (~1.1–1.25 GB), the 1-bit 8B is the better model.** It beats the 4-bit Qwen3-1.7B on GSM8K (+0.05, significant at n = 500), reverse-text (+0.04) and IFEval (+0.05, borderline), ties on MedMCQA, and with its adapters leads everywhere.
- **At equal parameters, the 4-bit 8B is the better model** on everything its base already does (GSM8K, MedMCQA, IFEval: +0.045 to +0.185 over the 1-bit base), and the self-distilled adapters do not close that gap.
- **Where a task needs a skill, an adapter beats parameters.** Reverse-text is a learnable transformation neither base performs; the 87 MB adapter takes the 1-bit model to 0.82, 0.61 above the 4-bit 8B, and is what puts the routed 1-bit library ahead on the macro average.
- **The public adapters do not help the 4-bit 8B under the Hub environments' contracts** — not on the official GGUF, where they are misapplied, and not on a correctly built one, where they run as trained but bring their own output format, stopping behaviour or objective (§3.6). Adapters trained on the environment's own rewards learn its contract by construction.

3.2 What an 87 MB adapter buys on the 1-bit base

Paired deltas over the same 200 examples and seeds (bootstrap 95% CI):

env	Q1 base	Q1 + our adapter	Δ	what the adapter learned
reverse-text	0.115	**0.817**	**+0.702** [+0.682, +0.722]	the task itself: the base reorders words instead of reversing characters and runs past the 384-token cap on half the items (mean 205 tokens, 49% truncated); with the adapter, 51 tokens, 0% truncated
gsm8k	0.845	0.875	+0.030 [−0.020, +0.075]	sharpening only — self-distillation on its own correct answers (loss 0.12 → 0.08); on 500 items +0.014 [−0.014, +0.044]
medmcqa	0.375	0.440	+0.065 [−0.005, +0.135]	real but modest: on 500 items 0.388 → 0.450, **+0.062 [+0.018, +0.108]**
ifeval	0.640	0.640	0.000 [−0.045, +0.040]	nothing measurable

The pattern is what one would predict for rejection-sampling fine-tuning: it can only amplify behaviour the base already samples. Where the base is far from the task (reverse-text) a few hundred gold examples move it by 0.7; where it already samples correct answers most of the time (gsm8k 91% of training samples correct) there is little left to amplify.

Training cost (RTX 5080 Laptop, 16 GB; the QLoRA job sat at the VRAM limit, so these are upper bounds):

adapter	data	sampling (Q1 server)	training	total
reverse-text	800 env gold reversals	—	7 min (150 steps)	7 min
gsm8k	757 of 1,600 samples correct	17 min	50 min (96 steps)	67 min
medmcqa	465 of 1,600	16 min	60 min (60 steps)	76 min
ifeval	479 of 1,600	23 min	80 min (60 steps)	103 min

3.3 Public Qwen3-8B adapters run on the 1-bit model

Bonsai shares Qwen3-8B's architecture, so rank-16 LoRAs trained on Qwen3-8B load onto the 1-bit weights unchanged. They change every output (0/200 texts identical to base on gsm8k and medmcqa, 33/200 on ifeval) but their effect is within noise at n = 200: gsm8k GRPO adapter +0.025 [−0.025, +0.075], MetaMathQA −0.005, MedMCQA adapter +0.045 [−0.030, +0.120], Tulu IF adapter −0.010 [−0.050, +0.025]. They neither break the 1-bit model nor reliably help it.

3.4 Adapters are narrow — and one is dangerous off-task

Every adapter on every env (1-bit base, 200 examples per cell; rows = env, columns = adapter):

env	no adapter	reverse-text	gsm8k	medmcqa	ifeval
reverse-text	0.115	**0.817**	0.131	0.133	0.135
gsm8k	0.845	**0.045**	**0.875**	0.850	0.875
medmcqa	0.375	**0.250**	0.370	**0.440**	0.340
ifeval	0.640	**0.330**	0.645	0.630	**0.640**

The three self-distilled adapters are close to harmless off-task. The reverse-text adapter — the one that learned a genuinely new behaviour — destroys the others (gsm8k 0.845 → 0.045: it starts reversing). A skill library on one base is therefore only as good as its router.

3.5 Routing the library: TRM, LDT and hybrids

End-to-end macro score (mean over the four envs of the score of the answer actually produced under the routed adapter; 800 prompts, TRM averaged over 5 seeds):

router	full prompt: routing acc	full: end-to-end	content only: routing acc	content: end-to-end
oracle (true env's adapter)	1.000	0.693	1.000	0.693
no adapter (base)	—	0.494	—	0.494
best single adapter for everything (medmcqa)	—	0.513	—	0.513
TRM (72k parameters)	0.998	**0.693**	0.865	**0.615**
LDT alone (lexical lattice; bottom → base)	0.305	0.506	0.305	0.507
hard-filter hybrid	0.935	0.693	0.683	0.553
soft hybrid / confidence arbitration / membrane (γ by CV)	= TRM	0.693	= TRM	0.615
membrane, abstain threshold tuned end-to-end (cross-fitted)	0.998	0.693	0.862	0.613
TRM + expected-utility over the cross-task matrix (cross-fitted)	0.643	0.684	0.570	0.603

Findings, which replicate the routing ablation of the TRM/LDT hybrid paper on a new problem:

1. **A 72k-parameter TRM is enough when the prompt carries the task**: it matches the oracle (0.693), turning four 87 MB adapters into +0.20 macro score over the base model, while no single adapter gets more than +0.02.
2. **Lexical lattice evidence is too noisy to filter with.** The LDT alone reaches bottom on most prompts (abstains 43–63%), and using it as a hard filter costs 0.06 end-to-end on content-only prompts — the same failure mode as the paper's routing benchmark (0.584 vs 0.815). As soft telemetry it is harmless and inert.
3. **Abstention does not rescue content-only routing here.** The errors are bare sentences (reverse-text items) routed to other adapters, and for those the base model is no better than a wrong adapter (0.115), so abstaining gains nothing; the costly error (a math problem sent to the reverse-text adapter) is rare (8/200). Expected-utility routing over the measured cross-task matrix is *worse* (0.603): off-task, three adapters are near-ties, and the decision chases noise in the estimated matrix.

3.6 Against the 4-bit 8B

Paired deltas, 1-bit Bonsai + our adapter minus 4-bit Qwen3-8B (our `Q4_K_M` from the exact `Qwen/Qwen3-8B` weights; 200 examples, bootstrap 95% CI):

env	Q1 + adapter (1.25 GB)	Q4 8B base (5.03 GB)	Δ vs Q4 base	Q4 8B + best online adapter	Δ vs Q4 + online
reverse-text	**0.817**	0.208	**+0.609** [+0.577, +0.644]	— (none exists)	—
gsm8k	0.875	**0.940**	**−0.065** [−0.110, −0.020]	0.860 (MetaMathQA)	+0.015 [−0.035, +0.070]
medmcqa	0.440	**0.560**	**−0.120** [−0.200, −0.030]	0.085 (MedMCQA LoRA; format)	+0.355 [+0.275, +0.440]
ifeval	0.640	**0.685**	−0.045 [−0.085, +0.000]	0.540 (Tulu IF)	+0.100 [+0.045, +0.155]

Read honestly, this is two results:

1. **On what the base models already know, the 4-bit 8B is better.** Bonsai at 1 bit trails Qwen3-8B at 4 bits by 4.5–18.5 points on math (0.845 vs 0.940), medical knowledge (0.375 vs 0.560) and instruction following (0.640 vs 0.685), and a self-distilled adapter narrows but does not close that gap (−0.065 on GSM8K and −0.120 on MedMCQA, both significant; against the official file at n = 500 the gaps are −0.070 and −0.068); IFEval −0.045 is borderline at n = 200.
2. **A trained skill beats the bigger model where neither base has it.** On reverse-text the 87 MB adapter puts the 1-bit model 0.61 above the 4-bit 8B — at a quarter of the memory.

What happened to the community adapters on the 4-bit model. On Qwen's official `Qwen3-8B-Q4_K_M.gguf` every public adapter lowered the score, two of them catastrophically: the MetaMathQA adapter stuttered ("glasses glasses") and fell to 0.425 on GSM8K (base 0.945; its author reports 0.840 with bf16 weights in vLLM), the Tulu IF adapter took IFEval from 0.68 to 0.375, and the GRPO GSM8K adapter scored 0.25. We traced the collapse to the *file*, not to 4 bits and not to llama.cpp (100 GSM8K items, greedy, MetaMathQA adapter unless noted; `results/diagnostics/`, `runs/peft/`):

Qwen3-8B weights the adapter is applied to	GSM8K
official `Q4_K_M` — `general.name` "Qwen3 8B **Awq Compatible** Instruct" (CUDA)	**0.37**
same file, CPU backend only (30 items)	0.47
official `Q8_0` — "Qwen3 8B Instruct"	0.87
our `Q4_K_M`, converted from the published `Qwen/Qwen3-8B` (base 0.95)	0.86
our `Q4_K_M` with f16 embeddings and output layer	0.88
our `Q4_0` / `Q5_K_M` / `Q6_K`	0.80 / 0.86 / 0.84
adapter merged into the weights first, then our `Q4_K_M` / `Q8_0`	0.86 / 0.89
transformers + PEFT on 4-bit NF4 (bitsandbytes), 60 items	0.85
author's report: bf16, vLLM	*0.84*

Every quantization built from the published weights — 4-bit K-quant, legacy 4-bit, 5- and 6-bit, with the adapter applied at runtime or merged in, in llama.cpp or in transformers — runs the adapter as its author reports. Only the official `Q4_K_M` fails, on both backends, and its metadata says why: it was converted from an "awq-compatible" checkpoint. Its RMSNorm weights differ from the published model's by 20–55% (attention and FFN norms, layers 0 and 17), while the official `Q8_0`'s are identical to ours. AWQ-style preparation moves per-channel scales between a norm and the projections it feeds: the network computes the same function, so the base model scores normally, but the projection matrices now sit in a rescaled input basis, and a LoRA trained against the original basis is applied to differently scaled activations. The bigger the trained delta, the bigger the damage — the long-trained MetaMathQA and Tulu adapters break, while our small self-distilled Bonsai adapters barely move the official file (GSM8K 0.940, MedMCQA 0.510, IFEval 0.660 vs base 0.945 / 0.515 / 0.680). (The GRPO adapter is also defective in itself: it never learned to stop and loops after its answer on every backend, NF4 included.)

The fair comparison therefore uses our own `Q4_K_M` built from the exact checkpoint the adapters were trained on (200 items, env sampling, as the main table):

public adapter	env	Q4 8B base → + adapter	paired Δ (95% CI)	official file	what goes wrong
MetaMathQA (tianzl66)	gsm8k	0.940 → 0.860	−0.080 [−0.130, −0.030]	0.425	nothing — it runs as its author reports (0.84) and is simply weaker than Qwen3-8B's own non-thinking GSM8K
GRPO GSM8K (Pradheep1647)	gsm8k	0.940 → 0.540	−0.400 [−0.470, −0.330]	0.25	never learned to stop: loops after `#### N` on every backend
MedMCQA QLoRA (vireshk)	medmcqa	0.560 → 0.085	−0.475 [−0.550, −0.400]	0.475	answers in its training format — a bare letter, then an explanation — which the env's `\boxed{}` parser rejects; by letter it is right 58.5% of the time (base 56.0%)
Tulu IF (tianzl66)	ifeval	0.685 → 0.540	−0.145 [−0.200, −0.090]	0.375	a weak adapter for this env at any precision

Our Bonsai adapters, by contrast, transfer *positively* to the correctly built Qwen3-8B `Q4_K_M` — the two models share an architecture and, it appears, enough weight geometry: reverse-text 0.208 → 0.282, MedMCQA 0.560 → 0.590, GSM8K 0.940 → 0.940, IFEval 0.685 → 0.690 (on the AWQ-rescaled official file the same adapters were neutral to harmful: 0.093 / 0.940 / 0.510 / 0.660).

So correcting the basis rescues the adapters' *intended* behaviour, and it still does not make them useful here. A community adapter brings the contract of whatever it was trained on — its answer format, its stopping behaviour, its objective — and a Hub environment scores against its own. The MedMCQA adapter is the

clearest case: it knows slightly more medicine than the base and scores a sixth as much. Adapters trained from the environment's reward (ours) learn the environment's contract by construction; that, not the bit width, is why they are the ones that help.

A deployment hazard the 1-bit route avoids by construction. Two GGUFs with the same model name and quantization type can hold different checkpoints; the difference is invisible in benchmark scores and only shows when an adapter is attached. Before attaching a LoRA to a downloaded GGUF, check `general.name` / `general.finetune` (or compare a norm tensor against the source weights), or convert the GGUF yourself from the checkpoint the adapter was trained on. Bonsai's training checkpoint (`Bonsai-8B-unpacked`) is an exact expansion of the served `Q1_0` weights, so adapters trained on it always target exactly the weights they run on.

3.7 Systems: memory and speed

llama-bench, PrismML fork build 10743, RTX 5080 Laptop (16 GB) / Ryzen AI 9 365:

model	weights	GPU decode (tg128)	GPU prefill (pp512)	CPU decode (tg32, -ngl 0)
Bonsai 8B Q1_0	**1.15 GB**	**200.6 tok/s**	4,949 tok/s	**17.7 tok/s**
Qwen3-8B Q4_K_M	5.02 GB	89.1 tok/s	3,137 tok/s	11.1 tok/s
Qwen3-1.7B Q4_K_M	1.10 GB	319.5 tok/s	16,362 tok/s	51.8 tok/s

At equal parameter count the 1-bit model decodes **2.25× faster on the GPU and 1.6× faster on the CPU** than the 4-bit one, from 4.4× less weight memory — decode is memory-bandwidth-bound, and the 1-bit weights are a quarter of the bytes.

The cost: adapters cannot be merged into 1-bit weights. A 4-bit model can absorb a LoRA by merging it into fp16 weights and re-quantizing; a 1-bit model cannot (the sum is no longer ±scale), so on Q1 every adapter runs as a separate low-rank product. Served throughput with one active adapter fell from ~390 to ~160–185 aggregate tok/s (6 concurrent requests). This is the price of the 1-bit footprint, and the reason the skill-library pattern — many adapters loaded, one active per request — is the natural deployment for Q1 rather than a merged specialist.

Deployment footprint (llama-server, one slot, 4k context; VRAM measured as the `nvidia-smi` delta when the server starts, RAM as its working set; `results/memory.json`):

server	VRAM	process RAM
Q1 Bonsai 8B	**1.87 GiB**	1.42 GiB
Q1 Bonsai 8B + all four adapters loaded	**2.37 GiB**	1.55 GiB
Q4 Qwen3-8B	5.23 GiB	4.78 GiB
Q4 Qwen3-8B + four adapters	5.73 GiB	4.91 GiB
Q4 Qwen3-1.7B	1.73 GiB	1.44 GiB

The whole four-skill 1-bit library, router included (72k parameters, negligible), runs in less VRAM than the 4-bit 8B base alone needs — within the budget of a 4 GB GPU, where the 4-bit 8B does not fit at all.

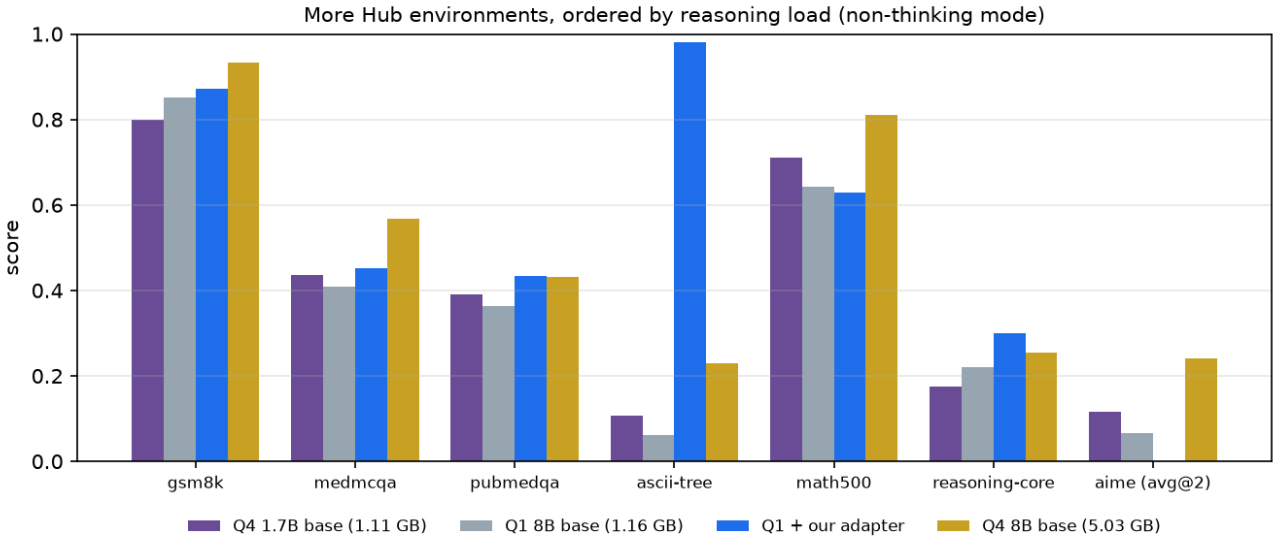
3.8 More environments, bigger samples: where one bit runs out

The four original environments are mostly behaviour and recall. To see where the 1-bit model's capacity runs out, we add five Hub environments that load reasoning progressively harder — biomedical evidence judgement (PubMedQA), a structured transformation (ascii-tree), competition math (MATH-500, AIME 2024+2025), and procedurally generated symbolic tasks (reasoning-core: 56 task families, grammars, logic, planning, automata, code tracing) — and rerun GSM8K on its full 1,319-item test set and MedMCQA on 1,000.

All models stay in non-thinking mode, as in §3.1; these are therefore *zero-scratchpad-budget* reasoning numbers, the hardest case for a small model.

env (metric)	n	Q4 1.7B (1.11 GB)	Q1 8B (1.16 GB)	Q1 + our adapter (1.25 GB)	Q4 8B (5.03 GB)	Q1 − Q4 1.7B	Q1 − Q4 8B
GSM8K (accuracy, full test)	1,319	0.798	0.851	0.871	**0.933**	+0.053 [+0.030, +0.076]	−0.081 [−0.099, −0.063]
MedMCQA (accuracy)	1,000	0.437	0.409	0.453	**0.567**	−0.028 [−0.062, +0.008]	−0.158 [−0.194, −0.121]
PubMedQA (accuracy)	500	0.390	0.364	**0.434** [1]	0.432	−0.026 [−0.074, +0.018]	−0.068 [−0.110, −0.028]
ascii-tree (tree similarity)	200	0.106	0.062	**0.981** [1]	0.230	−0.044 [−0.070, −0.019]	−0.168 [−0.202, −0.133]
MATH-500 (accuracy)	500	0.710	0.642	0.630 [2]	**0.810**	−0.068 [−0.108, −0.030]	−0.168 [−0.206, −0.130]
reasoning-core (task score)	500	0.175	0.220	**0.300** [1]	0.256	+0.045 [+0.010, +0.079]	−0.036 [−0.074, +0.002]
AIME 24+25 (avg@2 [pass@2])	60	0.117 [0.167]	0.067 [0.083]	—	**0.242 [0.300]**	−0.050 [−0.117, +0.017]	−0.175 [−0.267, −0.092]

Rows are ordered roughly by reasoning load. Deltas are paired over the same items and seeds, with bootstrap 95% CIs. [1] New phase N2 adapter (below). [2] Our GSM8K adapter applied to MATH-500. Every Q1 + adapter cell is 1.16 GB of base plus one 87 MB adapter.



Bigger samples sharpen two earlier calls. On the full GSM8K test (n = 1,319), Q1 beats the memory-matched 1.7B by +0.053 [+0.030, +0.076] (0.851 vs 0.798) and trails the 8B by 0.081 (0.933). Our RFT GSM8K adapter, "nothing measurable" at n = 200–500, is a small but real **+0.020 [+0.002, +0.038]**. On MedMCQA at n = 1,000, base Q1 sits below the 1.7B (−0.028 [−0.062, +0.008], not significant; 0.409 vs 0.437). Our MedMCQA adapter adds +0.044 [+0.015, +0.073], to 0.453, which puts the adapted 1-bit model above the 1.7B. The 8B leads it by 0.11 (0.567).

Where the 1-bit model holds up. On reasoning-core — short-answer symbolic tasks — Q1 beats the memory-matched 4-bit 1.7B (+0.045 [+0.010, +0.079]) and is within noise of the 4-bit 8B (−0.036 [−0.074, +0.002]). The per-family view has 56 families of 3–17 items each, too small to read individually, and it is

mixed. Q1 scores zero on a few families where the others do not (process inversion, metamath entailment), and leads on others (game moves, word problems). All three score zero on the same seven: logic derivation, controlled code execution, attribute grammars, dynamic programming, shift-reduce parsing, planning, belief tracking. At 0.18–0.26, this environment is weak for *all* three models, as expected of non-thinking 8B-class models on procedurally generated multi-step problems.

Where it runs out: multi-step math. The ordering that held on GSM8K (Q1 above the 1.7B) reverses on harder math: MATH-500 Q1 0.642 vs 1.7B 0.710 (−0.068 [−0.108, −0.030]) vs 8B 0.810; AIME avg@2 0.067 vs 0.117 vs 0.242 (pass@2 0.083 / 0.167 / 0.300). By MATH-500 difficulty level, Q1 is the lowest of the three at every level (level 1: 0.907 vs 0.977 / 0.977; level 5: 0.373 vs 0.433 / 0.582). It trails the 1.7B by a steady 0.06–0.09, and its gap to the 8B grows, not monotonically, from 0.07 at level 1 to 0.21 at level 5. Part of the gap is length. Q1 writes longer solutions and runs out of budget more often: 12.8% of its MATH-500 answers and 22.5% of its AIME answers hit the cap, against 9.6% / 10.0% for the 8B. On the answers that finish, Q1's MATH-500 accuracy is 0.736 (1.7B 0.792, 8B 0.894). One bit per weight costs the most where a derivation must stay exact over many steps. For this kind of task, the 4-bit 1.7B, not the 1-bit 8B, is the better use of ∼1.1 GB. An adapter does not change that: our GSM8K adapter does not transfer to MATH-500 (−0.012 [−0.052, +0.026]).

Weak for everyone, for format reasons: ascii-tree and PubMedQA. ascii-tree (render a prose directory description as an ASCII tree) is near the floor for all three base models (Q1 0.062, 1.7B 0.106, 8B 0.230), with 21–30% of outputs hitting the 768-token cap. It is not hard as reasoning: every node name of every gold tree appears in its prompt. The typical failure is copying the prompt's *style example* tree into the answer, then appending a garbled version of the real one. PubMedQA is a calibration failure more than a reasoning one: every model answers "maybe" on 59–65% of items against a 10% gold rate (Q1: 327/500 "maybe"; gold 276 yes / 173 no / 51 maybe). The env's system prompt primes this — its two worked examples are a "yes" and a "maybe" for "uncertain evidence", with no "no". Accuracy: Q1 0.364, 1.7B 0.390, 8B 0.432. Both are the shape of task that an environment-trained adapter fixed on reverse-text and MedMCQA, so phase N2 trains one for each. It also trains one for reasoning-core, on gold *answers only*, to ask whether a format-level adapter can buy anything on a reasoning task.

Phase N2: three more environment-trained adapters on the 1-bit model. Same recipe as §2 (rank 16, 87 MB, one laptop GPU, 37–89 minutes each):

env	training signal	Q1 base	Q1 + adapter	paired Δ [95% CI]	Q4 8B base
ascii-tree	env gold trees, 800 train rows	0.062	**0.981**	**+0.919 [+0.899, +0.936]**	0.230
reasoning-core	env gold *answers* (no derivation), 800 rows	0.220	**0.300**	+0.080 [+0.038, +0.119]	0.256
PubMedQA	rejection sampling: 400 train prompts × 4, 556 kept	0.364	**0.434**	+0.070 [+0.030, +0.110]	0.432

- **ascii-tree** is the reverse-text result again, on a second transformation skill. The 1-bit model plus an 87 MB adapter goes from the worst of the three to near-perfect, 4× the 4-bit 8B. We checked for leakage: no eval prompt appears in training, and 16 eval trees duplicate a training tree (the dataset is a mix). On the other 184 the adapter scores 0.979 (base 0.064, 8B 0.230). Every answer is derivable from its prompt, so this is a learned output discipline, not memorised trees.
- **reasoning-core**: an adapter that never saw a derivation lifts the score by 0.08, which puts the 1-bit model *above* the base 4-bit 8B. That is not a reasoning gain in the usual sense. 40 of 500 base answers lacked the `<answer>` tags (the adapter fixes all of them), but the gain is +0.069 [+0.027, +0.112] even on the 460 the base had already tagged. Answers shrink from ∼70 to ∼22 tokens. The model now answers directly, in each family's expected answer shape. By family it is a trade: large gains on metamath entailment, abduction and parsing derivations; losses on word problems, constraint satisfaction and probabilistic

outcomes. Those are the families where the base, still writing short derivations, was strongest (0.50–0.75). With 3–17 items per family, read these as directions, not estimates. Answer-shape adapters buy points on a procedurally generated benchmark; they do not make an 8B model reason.

- **PubMedQA**: rejection-sampled from the env's own (yes/no-only) training split, the adapter brings the 1-bit model level with the 4-bit 8B (0.434 vs 0.432). It does not cure the "maybe" habit (327 → 295 of 500); it gains on "yes" items (0.380 → 0.464) and "no" items (0.231 → 0.295).

4. What Q1 buys — and what it does not

- **Buy it for a skill library at a fixed small memory budget.** At ~1.2 GB the 1-bit 8B is the stronger base than the 4-bit 1.7B on recall and short-answer tasks (GSM8K, IFEval, reasoning-core; statistically tied on the two medical sets), though not on long derivations (below). It is the faster 8B (2.25× decode), and it takes cheap per-task adapters that a 72k-parameter router can dispatch. Where the deployment needs behaviours — formats, transformations, house styles — rather than more world knowledge, this is a better use of 1.2 GB than a smaller 4-bit model.
- **Do not buy it for knowledge.** If memory allows the 4-bit 8B, it is 4–16 points better on what both models already know, and self-distilled adapters do not close that gap.
- **Do not buy it for long exact derivations.** On MATH-500 and AIME the 1-bit 8B trails even the 4-bit 1.7B, its solutions run longer and hit the token cap more often, and a GSM8K adapter does not transfer. At ~1.1 GB, a multi-step math workload is better served by the smaller 4-bit model, and better still by a thinking-mode budget on a larger one. In non-thinking mode, every 8B-class model here is weak on AIME (≤ 0.24) and on procedurally generated reasoning (≤ 0.26).
- **Adapters buy output discipline, not reasoning.** The largest gains in this paper (+0.70 reverse-text, +0.92 ascii-tree) are transformation skills whose answer is fully in the prompt. An answer-only adapter gains +0.08 on reasoning-core by changing answer shape, while losing on the families where the base reasoned best.
- **Train adapters against the weights you serve — and against the environment you are scored by.** The 1-bit route gets the first for free (the unpacked checkpoint is the served model, bit for bit). On the 4-bit route, Qwen's own `Q4_K_M` GGUF is a different (AWQ-rescaled) checkpoint from the one community LoRAs are trained on, and silently breaks them; check `general.name` / `general.finetune` or convert the GGUF yourself. And even on the right checkpoint, a downloaded adapter answers in its own format and to its own objective: none of the four public adapters improved a Hub environment's score, while the environment-trained ones did.
- **Route; never apply a skill adapter blindly.** The one adapter that learned a new behaviour is catastrophic off-task (reverse-text adapter on GSM8K: 0.045). A tiny TRM routes full prompts perfectly; on bare content it reaches 86.5% and loses 0.08 of the oracle's macro score, so deployments should keep the task signal in the prompt.
- **Budget for unmerged adapters.** Q1 cannot fold an adapter into its weights; an active adapter roughly halved served throughput here.

5. Limitations

- **Eval sets.** 200 items for reverse-text and IFEval, and for the cross-adapter and online-adapter comparisons; 500–1,319 for the base-model comparisons (§3.8); 60 AIME problems × 2 samples. One sample per item elsewhere, and several differences are within noise and reported as such. reasoning-core has only 3–17 items per family, so per-family statements are directional.

- **Non-thinking mode only.** Sampling follows Qwen3's non-thinking recommendation for every model, with fixed token budgets (2,048 for MATH-500, 6,144 for AIME). The reasoning-heavy numbers (§3.8) are therefore lower bounds on what these models do with a scratchpad budget; thinking mode was not evaluated, and 10–22% of competition-math answers hit the budget.
- **Environment quirks are scored as-is.** PubMedQA's own prompt shows a "maybe" example and no "no" example, and every model over-answers "maybe". ascii-tree's prompts include a style-example tree that models copy. We score what the env scores and do not repair prompts. PrismML's own GSM8K/IFEval numbers come from a different harness and dataset version (their IFEval is the Google benchmark; ours is the Hub env's RLVR-IFEval holdout), so they are not directly comparable.
- **One adapter recipe.** Rejection-sampling fine-tuning from the base's own samples can only amplify what the base already does; RL against the environment reward (what the Prime Intellect stack is built for) or distillation from a stronger model would likely do more on GSM8K and IFEval.
- **Few public adapters.** Four were found that match these environments' shapes; their quality varies (the Tulu IF adapter is weak at any precision; the GRPO adapter never learned to stop). The official-GGUF mechanism is established for one adapter in depth (MetaMathQA: eight quantization / runtime / merge variants, CPU and CUDA, NF4 + PEFT) and checked by norm-weight comparison; the diagnostic grids used 100 items (60 for NF4, 30 for CPU).
- **One machine, noisy conditions.** An RTX 5080 Laptop shared with other jobs at times. Served throughput numbers were taken under that sharing and under a recurring software power cap (P-states P2–P8 at ~35–40 W even on AC power) that also caused Windows TDR GPU resets; the speed table uses isolated `llama-bench` runs instead. The PrismML fork crashed on Qwen3-1.7B, so that model was served by upstream llama.cpp (build b11269); interrupted runs were resumed per item, and partial files from the crashed binary were discarded. During the §3.8 runs the GPU was capped at 15 W (of 80 W). An NVIDIA driver bugcheck during a Modern Standby transition killed one adapter training mid-write; that adapter was retrained from scratch and every result file was checked for torn lines.
- **Router data.** Prompts from four environments are easy to separate when the instructions are kept; the content-only view is the harder, more realistic case, and the routers were not tested on out-of-distribution traffic. Routing (§3.5) covers the four original adapters; it was not retrained for the seven-adapter library of §3.8.

6. Reproducing

```
# environments venv (verifiers<0.4 + the nine Hub env packages) and the torch venv (QLoRA)
python harness/generate.py --env gsm8k --split eval --label q1-base --out results/gsm8k/q1-base
python harness/export_gold.py --env reverse-text --out runs/sft/reverse-text-gold.jsonl
python harness/generate.py --env gsm8k --split train --n 400 --k 4 --seed 1 --label q1-base-rft --
out runs/rft/gsm8k-q1-base
python harness/train_lora.py --env gsm8k --data runs/rft/gsm8k-q1-base.jsonl --min-score 1.0 --per-
prompt 2 --out <dir>
python harness/routers.py
python harness/make_tables.py
```

The phase scripts (`harness/phase{A,B,C,D,E,X,M,Y,Z,N,N2}.ps1`) record the exact server configurations (§3.8 is phases N and N2; `python harness/export_gold.py --env ascii-tree --n 800 --out ...` builds its gold-SFT data). All per-sample generations and scores are in `results/`; adapter weights are kept outside the repository.